

Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

What does inheritance in Java mean?



Learning Outcome:

Talk about principles, concepts, and characteristics of general-purpose computer programming language in spoken and written situations.

Ask your partner:

- What do you know about inheritance? What does it have to do with Java? Why?
- What does inheritance in Java have to do with object-oriented programming?

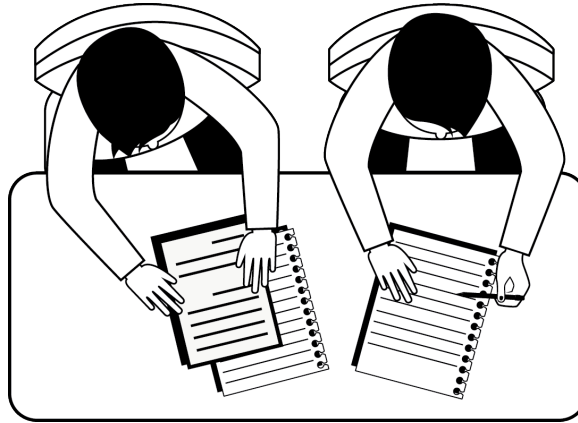
Can Do Statements:

- I can talk about what inheritance means in Java and its relation to object oriented programming.
- I can provide examples related to inheritance in Java.
- I can override in regards to inheritance in Java.



Let's Explore!

Jack needs a review on inheritance in Java!



Instructions: Read the text below.

- Jack:** Jena, I really need your help. In my Job, we didn't use to work with Java, but now my boss wants me to work on something called Java inheritance. You know that when we were in college I wouldn't pay much attention in our Java class.
- Jena:** Yes, and I remember you wouldn't go to that class very often. Don't worry, I have to work with Java repeatedly this semester and I can explain everything to you.
- Jack:** Thanks, can you tell me how to harness inheritance in Java? I know it has to do with object oriented programming.
- Jena:** Simple! Inherit in Java is when the program allows methods and variables that belong to one class to be accessible by another one without generating redundant codes. The main class that provides these items is called parent class, also known as superclass. The class that access these items is called child class or subclass.
- Jack:** That's why I wouldn't understand Java in college. Those concepts are easily misinterpreted.
- Jena:** Well, you didn't use to use examples to understand the theory, let me give you some. Imagine you create a parent class that represents an employee and a subclass that represents specifically a union employee; the union employee class is capable of using variables and methods from the parent class and creates its own but not vice versa.
- Jack:** It's clear now. One last question. Are child classes able to modify the methods they inherit from the parent classes?
- Jena:** Definitely! With a child class, you can override those methods. To override a data type, you just need to use the original method and revamp it with what you need. The only requirement is to use the super keyword in front of each method so, Java knows you're overriding.

**Let's Practice!**

Checking the override method in the inheritance in Java.

Instructions: Use the words from the Word Bank to complete each sentence.

Word Bank

repeatedly x2 / easily / badly / carefully

1. I need your help _____. I didn't understand the revamp concept.
2. I have to work with Java _____ this semester and I can explain everything to you.
3. Inheritance has to be done _____ with object oriented programming.
4. Method and variable concepts are _____ misinterpreted.
5. With a child class you can override those methods_____.

**Let's Practice!**

Defining How the Inheritance Works on Java

Instructions: Read the paragraph. Write the missing word(s) in the blank. Use: provides, is, allows, accesses

Inheritance in Java _____ when the program _____ methods and variables that belong to one class to be accessible by another one without generating redundant codes. The main class that _____ these items is called parent class, also known as superclass. And the class that _____ these items is called child class or subclass.





Extension Activity

More ways to understand inheritance

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about inheritance usages in java.
- Write notes and describe in a chart how to use it step by step.
- Describe in your own words and with some real examples how it works with methods and variables.
- Share with your partner and check extra questions.

Self-Assessment

Yes

No

- I look for information about inheritance usages in java.
- I describe in a chart how to use it step by step.
- I talk about inheritance in Java and its usages.

☐ ☐☐ ☐☐ ☐

Target Words

- **vice versa** (adverb): the other way around of something
e.g. The union employee class is capable of using variables and methods from the parent class and creates its own but not vice versa.
- **override** (verb): to modify the regular functioning of something
e.g. With a child class you can override those methods.
- **revamp** (verb): to put something up to date
e.g. You just need to use the original method and revamp it with what you need.
- **harness** (verb): to use something
e.g. Can you tell me how to harness inheritance in Java?
- **item** (noun): a piece of something that belongs to a whole
e.g. The main class that provides these items is called parent class.
- **redundant** (adjective): something that is repeated unnecessarily
e.g. One class to be accessible by another one without generating redundant codes.



Learning Points

Used To and Would - Negative Forms

You use **used to** and **would** to talk about past habits you don't have anymore.

Remember that there are two kinds of verbs: **state verbs** and **actions verbs**. The state verbs are the ones that don't describe an action, which means they are static. On the other hand, action verbs describe an action that lasts for a certain period of time.

That being said, there is a difference in use between **used to** and **would** to describe past habits. You can't use **would** to describe past events with state verbs, for example:

Used to:

When I was little, I used to live in Miami. **Correct!**

Would:

When I was little, I would live in Miami. **Incorrect!**

Now, let's study the negative structure of used to. You will need the auxiliary did + not and the main verb in present form, like this:

Subject + did + not (didn't) + verb in present + complement

Example:

I **didn't use to** know about inheritance in Java, but now I do.

And now, let's see the negative form of would. You will need the modal would + not and the main verb in present form, like this:

Subject + would + not + verb in present + complement

Example:

When I was a kid, I **wouldn't go** to the library to look information about object-oriented programming

Example:

*When I was a kid, I **wouldn't go** to the library to look information about object-oriented programming.*

Examples

- *In my Job we didn't use to work with Java.*
- *You didn't use to use examples to understand the theory.*





Learning Points

Adverbs of Manner

Adverbs of manner are the ones that indicate specific information of the verbs.

Adverbs of manner tell us the "how" of the actions. Usually, these adverbs are placed either after the verb or after the object in the sentence and they end with **-ly**.

For example:

Inheritance in Java means to use and re-use objects, methods, and variables **correctly**.

Remember that there are some adverbs that don't have the termination **-ly**. For example: well, hard, fast, etc.

Let's take a look at the next chart to see more **adverbs of manner**

Adverb of manner	Meaning
accidentally	when something happens by accident
badly	when something is very bad
carefully	when you do something and you are very careful or cautious
easily	when something is very easy
hurriedly	when something happens very fast
neatly	when something is very clean; is in order
punctually	very punctual
quietly	when something is very silent
repeatedly	When there is a lot of repetition

Examples

- *I need your help, badly.*
- *I have to work with Java repeatedly this semester*
- *Those concepts are easily misinterpreted.*

